

Deep Koopman Approach for Nonlinear Dynamics and Control of Tendon-Driven Continuum Robots

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I. INTRODUCTION

Abstract—Tendon-driven continuum robots (TDCRs) have received widespread attention in the medical domain due to their slender shape and flexibility. Modeling the dynamics of TDCRs involves continuum mechanics that result in nonlinear and computationally intensive models posing challenges for real-time control. In this work, we propose efficient, and control-oriented modeling of the nonlinear dynamics of TDCRs leveraging the deep Koopman approach. This method transforms the states of the system into an intrinsic nonlinear manifold, where the autonomous dynamics are approximated linearly, and the actuation input enters the system with a bilinear term. The proposed model captures the nonlinearities, including space-dependent variations in the system spectrum. Position control is implemented using a linear quadratic controller, leveraging the linear nature of the Koopman operator. The accuracy of the proposed method is experimentally validated using a dual-tendon robotic steerable catheter, achieving a position tracking error of 1.64 mm (SD = 0.74) for multi-sinusoidal, and 0.60 mm (SD = 0.30) for sinusoidal (0.05 Hz) target trajectories. The results demonstrate the potential for applying the proposed approach for real-time control of a broad range of TDCRs.

Index Terms—Deep Koopman, tendon-driven continuum robot (TDCR), nonlinear dynamics, real-time control, robotic catheter.

TENDON-DRIVEN continuum robots (TDCRs) have garnered substantial interest due to their small size and inherent flexibility, which are particularly advantageous in medical applications requiring precise navigation through complex, narrow anatomical pathways [1]. Unlike traditional robots, TDCRs are jointless and are built around a flexible backbone that provides structural integrity while being manipulated by one or more eccentrically positioned tendons to achieve deformations along their body [2]. A notable implementation of tendon-driven continuum devices in clinical settings is steerable cardiac catheters [3], [4], [5].

The flexible nature of continuum robots introduces nonlinearities leading to models that cannot be formulated in the conventional state space configurations [6]. A comprehensive dynamic modeling approach employs Cosserat rod theory, which captures deformations critical for robots with flexible backbones [7], [8]. Machine learning-based modeling techniques make it possible to develop models without physics-based formulations, while capturing the comprehensive dynamics of TDCRs [9], including hysteresis [10]. Despite extensive research dedicated to the development of accurate modeling of TDCRs, significant challenges persist, especially in the context of real-time control. The Cosserat rod-based models lack closed-form solutions and suffer from inconsistent computation times, while the closed-box nature of conventional machine learning-based models makes them not amenable to conventional model-based control design techniques. This necessitates nonlinear controllers, posing challenges for real-time computation essential for their implementation.

Koopman operator theory [11] introduces control-oriented models for nonlinear dynamical systems by transforming them onto an infinite-dimensional nonlinear manifold (a.k.a. *lifted space*), where the dynamics becomes globally linear. Since an infinite-dimensional space is not practically applicable, various methods have been proposed to provide a projected approximation. Dynamic mode decomposition (DMD) provides a linear approximation of the Koopman operator [12]. However, it fails to capture nonlinear lifting functions and thus cannot span a Koopman-invariant subspace. The extended DMD (EDMD) technique improves upon DMD by incorporating nonlinear measurements for the lifting functions [13]. Nevertheless, EDMD faces two primary challenges: (i) the selection of basis functions is manual and relies on intuition [14]; and (ii) EDMD cannot

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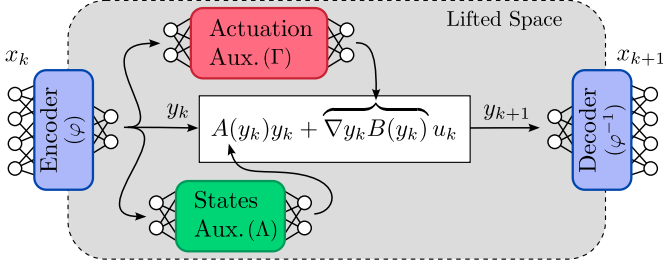


Fig. 1. Diagram of the continuous spectrum deep Koopman architecture, featuring an autoencoder and two auxiliary networks for modeling space-dependent parameters.

accurately map systems that are not globally topologically conjugate into a finite-dimensional linear system. Consequently, these systems cannot be represented by a linear Koopman subspace [15], [16].

Deep learning has proven effective for data-driven discovery and representation of appropriate lifting functions [14]. Studies have shown that using a deep learning approach to learn lifting functions outperforms EDMD in reconstruction accuracy with a smaller number of lifting functions [17]. However, this method falls short in capturing the dynamics of systems with a continuous frequency spectrum [18]. Several approaches have been proposed to capture continuous spectrum properties of system dynamics, including delay embedding and nonlinear spectral analysis [19], [20]. An intuitive method proposed in [21] demonstrated that incorporating the parameter dependency of the system's eigenvalues into the states effectively represents the continuous spectrum characteristics.

By leveraging the globally linear Koopman operator, conventional linear controllers can be implemented in the lifted space, with control actions subsequently transformed into the original state space [22]. This approach has shown promising results for the control of soft robots [23] and for modeling and control of the twist (excluding bending) of a flexible ureterscope [24]. Deep learning has also been used to identify Koopman models for controlling an inverted pendulum in [25], a flexible inverted pendulum in [26], and Franka and KUKA robotic arms in [27] and [28], respectively.

In this work, we take inspiration from the work reported in [21] and [22] to propose a continuous spectrum deep Koopman (CSDK) approach with control input to model the dynamics of TDCRs. To the best of our knowledge, an application of a deep Koopman approach has not so far been investigated for any type of TDCRs, including robotic catheters. We derive the structure of the model based on the properties of the catheter dynamics. In contrast to state-of-the-art approaches for Koopman-based control that utilize high-dimensional models with time-invariant dynamics, we propose to incorporate the variations of the system spectrum, using auxiliary neural networks, resulting in a low-rank model, as shown in Fig. 1. Additionally, we employ an autoencoder structure to identify Koopman eigenfunctions and reconstruct the predicted system state. A real-time closed-loop controller is implemented in the lifted space and the performance of the proposed controller is experimentally validated using a robotic tendon-driven catheter.

II. BACKGROUND

Consider an unforced system governed by the following differential equation:

$$\dot{x}(t) = f(x(t)) \quad (1)$$

where $x(t) \in \mathcal{M} \subset \mathbb{R}^n$ represents the state of the system at time $t \geq 0$, $\mathcal{M}$ is a differentiable manifold representing the finite-dimensional state space, and $f: \mathcal{M} \rightarrow \mathcal{M}$ is a continuously differentiable function that describes the dynamics of the system expressed in discrete time as:

$$x_{k+1} = F(x_k) \quad (2)$$

where x_k denotes the state of the system at the discrete time step k , corresponding to $x(kT)$, where T is the constant sample time. Here, $F: \mathcal{M} \rightarrow \mathcal{M}$ is the flow map, which advances the state of the system in time, defined by:

$$F(x(t)) = x(t_0) + \int_{t_0}^{t_0+t} f(x(\tau)) d\tau \quad (3)$$

According to Koopman spectral theory [11], the evolution of the nonlinear system F can be described by a linear evolution of observables (a.k.a. *measurables*) in an infinite-dimensional Hamiltonian function space, $\mathcal{F} \subset \mathbb{R}^p$. The function space, $\mathcal{F}$, is composed of all continuous real-valued observables $g: \mathcal{M} \rightarrow \mathcal{F}$, where g is a nonlinear function of the state x . In the space $\mathcal{F}$ the evolution of the dynamics of the system is characterized by the infinite-dimensional linear Koopman operator $\mathcal{K}_t: \mathcal{F} \rightarrow \mathcal{F}$

$$\mathcal{K}_t g = g \circ F \quad (4)$$

where $\circ$ denotes function composition. So we have

$$\frac{d}{dt} g(x) = \nabla_x g(x) \dot{x} = \nabla_x g(x) \cdot f(x) = \mathcal{K} g(x) \quad (5)$$

where $\mathcal{K}$ is the infinitesimal generator of the one-parameter family of Koopman operator $\mathcal{K}_t$ [15].

In the discrete-time representation of (4), the observables advance linearly in the function space $\mathcal{F}$ as follows:

$$g(x_{k+1}) = \mathcal{K}_t g(x_k) \quad (6)$$

Despite the linearity of the Koopman operator, its infinite dimensionality makes it computationally intractable. In systems characterized by a single isolated equilibrium point, a reduced-order model can be realized through a truncated approximation of the infinite-dimensional space $\mathcal{F}$ [15]. This allows for obtaining an applied finite-dimensional representation of the Koopman operator $\mathcal{K}$ by restricting it to an invariant subspace $\mathcal{H} \subset \mathcal{F}$ [15]. The subspace $\mathcal{H}$ can be defined by a set of eigenfunctions of the Koopman operator, where each eigenfunction (a.k.a. *lifting function*) denoted as $\varphi(x): \mathcal{M} \rightarrow \mathcal{H}$, is associated with an eigenvalue λ . The applied Koopman operator can be represented as a matrix when restricted to an invariant subspace spanned by eigenfunctions. Thus we have,

$$\frac{d}{dt} \varphi(x) = \lambda \varphi(x) = A \varphi(x) \quad (7)$$

where A is the Koopman operator, which is the same as the state matrix in linear systems.

III. METHODS

A. Extension to Control-Affine Dynamical Systems

Assuming a control-affine system, by adding m inputs that linearly enters the autonomous system (1), we obtain a dynamical system of the form

$$\dot{x}(t) = f(x(t)) + \sum_{i=1}^m b_i(x(t)) u_i(t) \quad (8)$$

where $f(x)$ is a continuously differentiable function representing the autonomous part of the dynamics, u_i are the control inputs, $b_i(x)$ are smooth vector fields that modulate the influence of each control input.

Now by lifting the states of the system in (8) using the lifting function $\varphi(x)$ associated with the dynamics of the unforced system (1) and applying the chain rule we have

$$\begin{aligned} \frac{d}{dt}\varphi(x) &= \nabla_x \varphi(x) \left(f(x) + \sum_{i=1}^m b_i(x) u_i \right) \\ &= A(\varphi(x))\varphi(x) + \nabla_x \varphi(x) \sum_{i=1}^m b_i(x) u_i \\ &= A(\varphi(x))\varphi(x) + \nabla_x \varphi(x) B u \end{aligned} \quad (9)$$

Thus, the dynamics of a nonlinear control-affine system, when represented in the lifted space $\mathcal{H}$, are consists of a linear part corresponding to the autonomous system and a bilinear part associated with the control input [22].

Without loss of generality, let us assume the dynamics of the controlled system in the lifted space $\mathcal{H}$ are given by the following equation:

$$\dot{y} = A(y)y + B_\phi(y)u \quad (10)$$

where $y = \varphi(x)$, $B_\phi(y) = \nabla_y \varphi^{-1}(y)B = \nabla_x \varphi(x)B$, and A and B are lifted states dependent.

In order to derive the equivalent discrete-time system, the non-homogeneous term is isolated, and subsequently both sides are premultiplied by the matrix exponential e^{-At} , yielding:

$$\frac{d}{dt} [e^{-At}y] = e^{-At}B_\phi(y)u \quad (11)$$

Assuming that $B_\phi(y)$ remains approximately constant and u is held constant over the sampling interval T via a zero-order hold, by integrating both sides and isolating $y((k+1)T)$, we have:

$$y((k+1)T) = e^{AT}y(kT) + \left(\int_0^T e^{A\tau} d\tau \right) B_\phi(y(kT))u(kT) \quad (12)$$

which leads to the compact form:

$$y_{k+1} = G(T)y_k + H(T)u_k \quad (13)$$

where the lifted states transition matrix $G(T)$ and the input actuation matrix $H(T)$ are given by:

$$G(T) = e^{AT}, \quad H(T) = \left(\int_0^T e^{A\tau} d\tau \right) B_\phi \quad (14)$$

It should be noted that both G and H are lifted space-dependent matrices slowly varying with the lifted states. If matrix A is non-singular, $H(T)$ simplifies to the following expression, as

derived from standard properties of matrix exponentials [29] [Page 313]:

$$H(T) = A^{-1}(e^{AT} - I)B_\phi = A^{-1}(e^{AT} - I)\nabla_x y B \quad (15)$$

B. Accounting for Continuous Spectrum

Although Koopman theory achieves global linearization of dynamics, it is limited to systems with discrete spectra. In contrast, as will be shown in Section V, physical systems, such as TCDRs display a broad range of continuous spectra, where an exact eigenfunction may not exist [18]. Variations in the system's spectrum can be approximated by utilizing a high-dimensional Koopman operator, resulting in a dynamical representation in a lifted space with dimensions far exceeding the original state space. Incorporating the parameter dependency of the eigenvalues of the Koopman operator can represent the continuous spectrum characteristics of the system using a parsimonious model and thereby match the intrinsic low-order dynamics [21]. Thus, instead of identifying a constant matrix A , an auxiliary network $\Lambda(\varphi(x))$ is used to parameterize the eigenvalues $\lambda = \mu \pm i\omega$ and A is constructed accordingly.

Remark: The Koopman operator A can be constructed with any of the canonical forms. However, since we will use deep learning for identifying the eigenfunctions, a structure that leads to eigenfunctions in the same numerical range is preferable. Thus, we selected the Jordan canonical form to obtain a block diagonal structure for A . For each real eigenvalue, the associated diagonal element is equal to the eigenvalue, and for each complex-conjugate pair of eigenvalues $\mu \pm i\omega$, in a continuous-time system, a 2×2 block is of the form:

$$J(\lambda) = \begin{bmatrix} \mu & -\omega \\ \omega & \mu \end{bmatrix} \quad (16)$$

When λ does not strongly depend on y or varies slowly in the sampling interval T , the discrete-time equivalent of the continuous-time Jordan block (16) is obtained by calculating the matrix exponential of $J(\lambda)$.

$$e^{J(\lambda)T} = e^{\mu T} \begin{bmatrix} \cos(\omega T) & -\sin(\omega T) \\ \sin(\omega T) & \cos(\omega T) \end{bmatrix} \quad (17)$$

By developing a nonlinear model of a TDCR based on the Cosserat rod theory [7], [8] and assuming the system's states consist only of the position and velocity of points along the backbone, it can be proven that the system is control-affine. Consequently, a TDCR can be represented by incorporating its continuous spectrum characteristics, as shown in (10) for continuous time or (13) for discrete time.

C. Network Architectures

In this work, we use deep learning for model identification, utilizing an autoencoder to identify $\varphi(x)$ and $\varphi^{-1}(x)$. We also use two networks to learn the dependency of B_ϕ and the spectrum of A on the states, as shown in Fig. 1. The matrix B can be determined using an auxiliary input network, and B_ϕ can be calculated utilizing the gradient of the encoder network. But, we directly determine B_ϕ to reduce the computational complexity of calculating the Jacobian of the encoder.

TABLE I
NETWORKS ARCHITECTURE

Network	Layers	Nodes	Act.	Input	Output
Encoder φ	4	128	ReLU	4	2
Decoder φ^{-1}	4	128	ReLU	2	4
State Auxiliary Λ	3	30	ReLU	2	2
Actuation Auxiliary Γ	3	30	ReLU	2	2

Multi-layer perceptrons were utilized for the encoder, decoder, state auxiliary, and actuation auxiliary functions. The encoder input size corresponds to the dimensions of the system state. The dimensionality of the lifted state space is determined by the nature and number of the eigenvalues. In our exploratory analysis, we evaluated models with different configurations, including a pair of complex-conjugate eigenvalues, a pair of complex-conjugate with an additional real eigenvalue, and two pairs of complex-conjugate eigenvalues. The analysis indicated that configurations with more than a pair of complex-conjugate eigenvalues did not vary significantly across the space, suggesting redundancy in the additional eigenvalues for a planar TDCR. Consequently, we incorporated a pair of complex-conjugate eigenvalues, mirroring a second-order system within the lifted space. The detailed architecture of each network component is presented in Table I.

D. Training

We employ a composite loss function consisting of a weighted sum of distinct components, including reconstruction loss, linearity loss, and prediction loss, incorporating all components of the model [21].

$$\mathcal{L} = \mathcal{L}_{lin} + \alpha_1(\mathcal{L}_{pred} + \mathcal{L}_{rec}) \quad (18)$$

$$\mathcal{L}_{lin} = \sum_{i=1}^n \|\varphi(x_{i+1}) - y_{i+1}\|_2 \quad (19)$$

$$\mathcal{L}_{rec} = \|x_1 - \varphi^{-1}(y_1)\|_2 \quad (20)$$

$$\mathcal{L}_{pred} = \sum_{i=1}^n \|x_{i+1} - \varphi^{-1}(y_{i+1})\|_2 \quad (21)$$

where n is the number of samples, α_1 is set to 0.1, and

$$\begin{cases} y_1 = \varphi(x_1), \\ y_{i+1} = A(y_i)y_i + B_\phi(y_i)u_i, \text{ for } i \geq 1. \end{cases} \quad (22)$$

The model was implemented in PyTorch. The model weights were initialized with a truncated normal distribution. The Adam optimizer [30] was used for optimization, with a learning rate of 1×10^{-3} and decay rate of 0.999. The training was conducted over 4000 epochs, with a batch size of 400.

IV. LQ CONTROL

In this work, we employ an infinite-horizon linear quadratic (LQ) controller. Leveraging the inherent linearization of the Koopman operator, the controller is implemented in the lifted space to stabilize the lifted state. To address the absence of integral action in the standard LQ controller formulation that might cause biased tracking (steady-state errors), the control signal

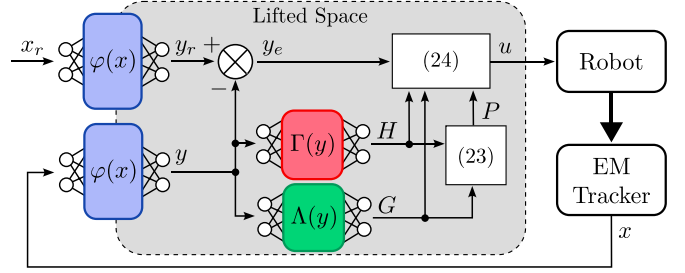


Fig. 2. Block diagram of the proposed deep Koopman LQ control system.

increment is penalized. The state vector is augmented to include the previous control input, enabling the control increment to be explicitly considered in the cost function. Consequently, the augmented system dynamics are given by:

$$\tilde{y}_{k+1} = \tilde{G}\tilde{y}_k + \tilde{H}\Delta u_k. \quad (23)$$

where

$$\tilde{G} = \begin{bmatrix} G & H \\ 0 & I \end{bmatrix}, \quad \tilde{H} = \begin{bmatrix} H \\ I \end{bmatrix}, \quad \tilde{y}_k = \begin{bmatrix} \varphi(x_k) \\ u_{k-1} \end{bmatrix} \quad (24)$$

This results in a quadratic tracking cost function given by

$$J = \sum_{k=1}^N \tilde{y}_{ek}^\top \tilde{Q}_\varphi \tilde{y}_{ek} + \Delta u_k^\top R \Delta u_k \quad (25)$$

with

$$\tilde{y}_{ek} = \begin{bmatrix} \varphi(x_k) \\ u_{k-1} \end{bmatrix} - \begin{bmatrix} \varphi(x_r) \\ u_{k-2} \end{bmatrix}, \quad \tilde{Q}_\varphi = \begin{bmatrix} Q_\varphi & 0 \\ 0 & 0 \end{bmatrix}. \quad (26)$$

where Q_φ is a positive definite matrix that denotes weighting on the lifted state, R is a positive-definite matrix that denotes the weighting on the input controls, and $\Delta u_k = u_k - u_{k-1}$ is the control input increment. With the dynamics expressed in their linear form, the solution to the optimal control (25) is simplified to solving the discrete algebraic Riccati equation:

$$\tilde{P} = \tilde{G}^\top \tilde{P} \tilde{G} - \tilde{G}^\top \tilde{P} \tilde{H} (\tilde{H}^\top \tilde{P} \tilde{H} + R)^{-1} \tilde{H}^\top \tilde{P} \tilde{G} + \tilde{Q} \quad (27)$$

The optimal state feedback control input is then given by:

$$u_k = u_{k-1} + K \tilde{y}_{ek} \quad (28)$$

where

$$K = -(\tilde{H}^\top \tilde{P} \tilde{H} + R)^{-1} \tilde{H}^\top \tilde{P} \tilde{G}. \quad (29)$$

As the state transition and actuation matrices G and H are state-dependent we update the matrices G and H at each control loop (see Fig. 2) and compute the control gains online, a method that can be interpreted as gain scheduling.

V. EXPERIMENT: TENDON-DRIVEN CONTINUUM ROBOT

A. Setup Description

A *ThermoCool SmartTouchTM SF bi-directional navigation catheter* (Biosense Webster, CA) was adapted to develop a robotic actuated tendon-driven system. The handle of the catheter was dismantled to attach the tendons to our custom-designed actuation unit. We also limit the bending section to 55 mm to equalize the length of the bending sections on both sides. The tendons are actuated by custom-made linear actuation units, as depicted in Fig. 3(a) [31]. All joints are

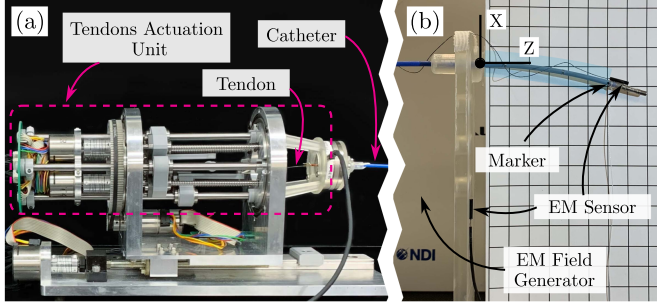


Fig. 3. Experimental setup. (a) Actuation unit. (b) Distal end of the catheter.

powered by brushless DC motors (Faulhaber, Germany). The control architecture was implemented using the Robot Operating System version 2 (ROS2). A dedicated ROS2 node in C++ was developed to enable synchronous communication with the motion controllers via the CANopen protocol. Position tracking of the distal end of the catheter within the frame is achieved using electromagnetic (EM) tracking sensors. Because the focus of this work is on the bending section, the ablation probe is excluded from this analysis, and only the transformation of the terminal point of the bending section (marked by a black indicator in Fig. 3(b)) is measured by the EM sensors. Footage from the final experiment with the robot can be found in the supplementary video file.¹

B. Dataset Generation

To achieve accurate model identification and capture the dynamics of the system, a comprehensive dataset encompassing the entire state space and the dynamic modes is essential. The ideal way to excite all dynamic modes of the TDCR for dataset generation is through random impulse actuation of the tendons. However, due to the spectral nature of the bending section of the catheter, the velocity of the tip exceeds the measurable limits of the EM tracking system, particularly during fast transitional dynamics. This limitation prevents the generation of a sufficiently rich training dataset capable of capturing the high-speed dynamics of the system directly from the experimental setup.

To address this issue, we calibrated a classical Cosserat rod model based on the setup parameters, which was subsequently employed to generate the training dataset for the deep Koopman model [7], [8]. Initially, parameters such as the length L , mass m of the bending section, and mass of the probe m_e were directly measured. Other parameters that could not be directly measured were calibrated through a two-stage optimization process. Experimental data was first collected by releasing a 20-gram weight from the tip of the catheter while the tendons were disconnected from the actuation unit. Since parts of the motion of the catheter tip exceeded the measurable limit of the EM tracking system, we used a 240 Hz camera and extracted the motion offline from the recorded video. This data was used to calibrate the catheter sheath parameters, including shear modulus E , Poisson's ratio ν , equivalent sheath radius r , and viscous damping coefficients

TABLE II
MEASURED AND CALIBRATED PARAMETERS OF THE COSSERAT MODEL

P	V	P	V	P	V
κ	$4.66 \times 10^{-4} \text{ N}^{-1}$	ν	0.385	L	55 mm
γ_{bt}	$1.50 \times 10^{-6} \text{ Nm}^2\text{s}$	r_{ten}	1.11 mm	m	0.516 g
γ_{se}	$7.50 \times 10^{-5} \text{ Ns}$	r	0.40 mm	m_e	0.338 g
E	15.47 GPa	—	—	—	—

for bending/torsion γ_{bt} and shear/extension γ_{se} . Additional experimental data was acquired using the EM tracker with tendon actuation to calibrate the tendon parameters, including tendon offset r_{ten} , and compliance κ . The calibrated Cosserat model resulted in the position accuracy of 1.62 mm (SD = 1.01) with the measured and calibrated parameters shown in Table II (see [32] for further details).

We defined the system states as the position and velocity of the distal end of the bending section in the actuation plane, denoted as $(x, \dot{x}, z, \dot{z})$, with tendon displacement used as the input. To analyze the dynamics of the autonomous and non-autonomous parts of the model separately, two datasets were created for the unforced and forced systems.

For the unforced dataset, tendon actuation was selected uniformly randomly within the range of $(0, 6]$ mm, covering the entire actuation range of the catheter. After waiting for the transition to stabilize, the actuation was followed by a retraction to zero at a velocity of 0.10 m/s, the robot's maximum applicable velocity. For the forced dataset, the tendon was actuated with a ramp signal at the maximum velocity of the robot to a random value, uniformly distributed within the actuation range. After each ramp signal, the tendon was held still for half a second to allow the transition dynamics to damp out. This cycle was repeated to ensure extensive coverage of the entire state space. Since the proposed model is intended for use in closed-loop control, where the nature of the control signal is unpredictable, the dataset was generated to include the highest-frequency modes of the system that can be excited by the robot. Subsequent data sets comprising 40 samples each were randomly selected, resulting in 5200 training and 800 validation sequences for both scenarios. Both datasets were sampled at 2 ms intervals, accumulating 1600 seconds of unforced and 1200 seconds of forced data.

VI. RESULTS AND DISCUSSION

A. Modeling

In order to evaluate the performance and dynamics of the unforced deep Koopman models, we conducted a comparative analysis using 200 test trajectories, each originating from a randomly selected initial condition. Fig. 4 shows the phase portraits of the states. Fig. 5 shows μ and ω of the eigenvalues (continuous time) computed by the states auxiliary network $\Lambda(y)$, denoting the decay rate and the oscillation frequency, respectively ($\mu = \zeta\omega_n$ and $\omega = \omega_d$, where ζ is the damping rate, ω_n is the undamped natural frequency, and ω_d is the damped frequency in a second-order system). As seen, $\zeta\omega_n$ and ω_d exhibit smooth variations, with no significant fluctuations observed. The negative decay rate confirms the stability of the

¹[Online]. Available: <https://youtu.be/OxP7nIUO6wc>

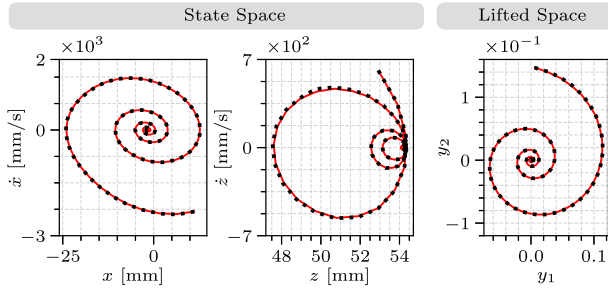


Fig. 4. Predictions from the autonomous CSDK model (black dots) and the Cosserat model (solid red lines).

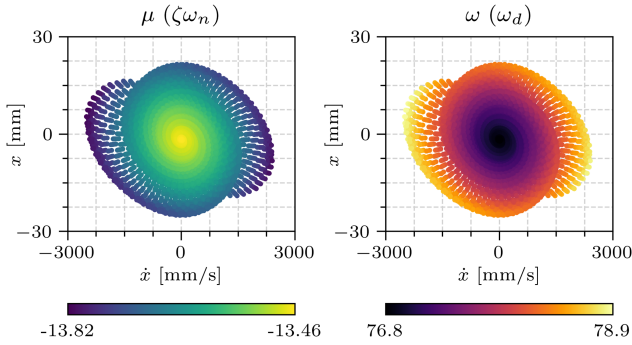


Fig. 5. Real (μ) and imaginary (ω) components of the eigenvalues of the continuous-time unforced CSDK model are presented as functions of the x and $\dot{x}$ state. μ represents the decay rate of the oscillations, and ω indicates the frequency of oscillation.

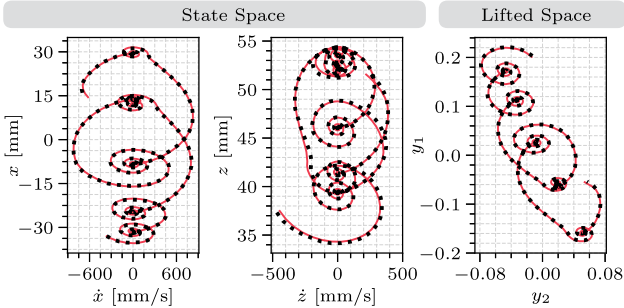


Fig. 6. Predictions based on the forced CSDK model (black dots) and the Cosserat model (solid red lines) for one of the test trajectories.

system and $\omega_d \approx 77$ rds (12.6 Hz) indicates the frequency of the system oscillations.

In order to validate the forced CSDK model, we employed a similar approach but with 60 pseudo-random near-step signals with periods of 0.5 seconds. Fig. 6 displays the phase portraits of the states predicted by the CSDK model alongside those from the Cosserat rod model for one of the test trajectories. As observed, the CSDK model aligns closely with the Cosserat rod model during both steady-state and transitional dynamics.

Fig. 7 shows the variation of the eigenvalues within both the x - $\dot{x}$ plane and the x - z plane. As seen, the system maintains stability, and the eigenvalues exhibit smooth transitions across the space. However, these variations are notably more pronounced compared to those for the unforced system. This is attributed to the actuation of the tendons, which influence the stiffness of the bending section. The heatmap of μ indicates that the oscillations

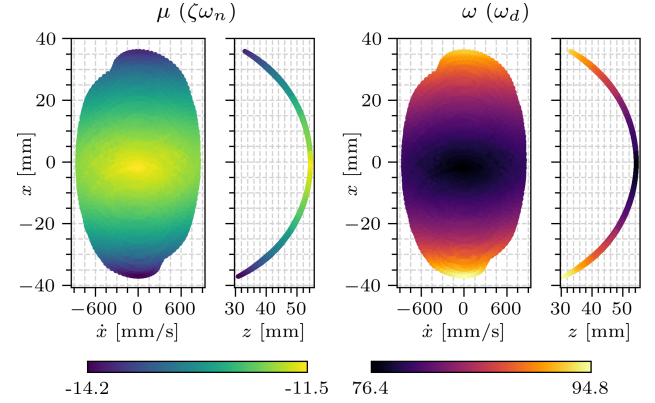


Fig. 7. Real (μ) and imaginary (ω) components of the eigenvalues of the continuous-time forced CSDK model as a function of the states; μ shows the decay rate and ω shows the frequency of oscillations.

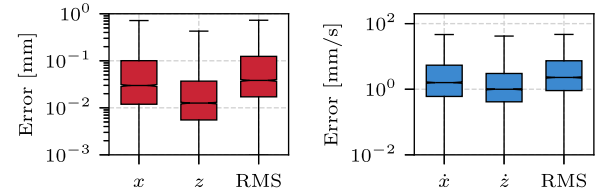


Fig. 8. Distribution of the position (red) and velocity (blue) prediction errors.

decay approximately 23% faster when the catheter is bent in one direction compared to its relaxed state. Similarly, the heatmap of ω demonstrates an increase in oscillation frequency by nearly 23% upon bending of the catheter. While the ζ remains relatively constant at approximately 0.4, the primary variations correspond to the ω_n .

Fig. 8 illustrates the distribution of prediction errors across all test trajectories (464 seconds). The root mean square of the position and velocity prediction errors measured were 0.08 mm (SD = 0.10), and 5.75 mm/s (SD = 7.50), respectively.

B. Comparison With Other Deep Koopman Models

An extensive evaluation was conducted to compare the accuracy of the proposed CSDK method with other deep Koopman approaches incorporating control input from the literature. The methods compared include: (1) direct input deep Koopman (DIDK) [27], [28], which represents the system as $\varphi(x_{k+1}) = G\varphi(x_k) + Hu_k$; (2) control affine deep Koopman (CADK), formulated as $\varphi(x_{k+1}) = G\varphi(x_k) + Hg(x_k)u_k$; and (3) nonlinear input deep Koopman (NIDK) [25], formulated as $\varphi(x_{k+1}) = G\varphi(x_k) + Hg(x_k, u_k)u_k$. In all cases, models are time-invariant, and the first elements of the lifting function φ correspond to the unity transformation of the system states, thereby eliminating the need for a decoder to reconstruct the predicted system states. Fig. 9 shows the results of the prediction errors across a range of lifted state dimensions. As observed, the accuracy of the proposed CSDK model is remarkably better than that of DIDK, CADK, and NIDK, except for NIDK when the lifted dimensions exceed 50.

In DIDK, the relationship between the input and the lifted states is assumed to be linear, which conflicts with the bilinear nature of the TDCR. This mismatch compromises the accuracy

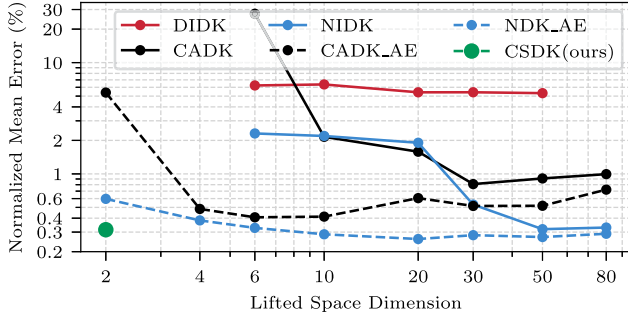


Fig. 9. Normalized mean error of various deep Koopman models.

in capturing the dynamics. Koopman theory is grounded in finding eigenfunctions that evolve linearly. While CADK and NIDK capture the nonlinear, control-affine dynamics of the TDCR, the extension of the lifted state (which is supposed to capture linear dynamics) with the robot states might not naturally align with linear property, leading to reduced performance. Additionally, these methods neglect the continuous spectrum nature of the system by assuming constant eigenvalues, leading to a higher demand for larger lifted state dimensions to compensate for it (see [21]).

To investigate this matter further, we developed two additional modified structures: CADK with autoencoder (CADK-AE) and NIDK with autoencoder (NIDK-AE). In these structures, the system states were removed from the lifted states, and an autoencoder approach was used to reconstruct the system state. As observed, the accuracy of both methods significantly improved in comparison to the original structure, particularly when the dimension of the lifted states was small. CSDK model, with only two-dimensional lifted states, outperforms CADK-AE across all lifted state dimensions. NIDK-AE achieved slightly better accuracy than CSDK when the lifted state dimension exceeds six. This highlights the critical role of integrating the continuous spectrum nature of the system in CSDK, enabling it to achieve optimal performance with the minimal possible lifted state dimension.

The CSDK model also offers the advantage of consistently lower computation times. Simulations for both models were conducted on an Intel i7-10750H with CPU affinity settings adjusted to allocate a dedicated core for each model. Predictions using the deep Koopman model were executed on a CPU utilizing the libtorch C++ library. Similarly, the Cosserat rod model was implemented and optimized in C++. The evaluations yielded a computation time of 0.44 ms (SD = 0.14) per time step for the deep Koopman model, significantly outperforming the Cosserat rod model, which required 3.26 ms (SD = 34.86) per time step. The computation time for the Cosserat rod model exhibited sensitivity to the input dynamics, with high-frequency inputs necessitating extended computation times to converge.

C. Closed-Loop Control

The closed-loop LQ control approach, discussed in Section IV, was implemented on experimental setups. The real-time control system was executed in ROS2, with the control loop operating at a frequency of 250 Hz. Two different signals were

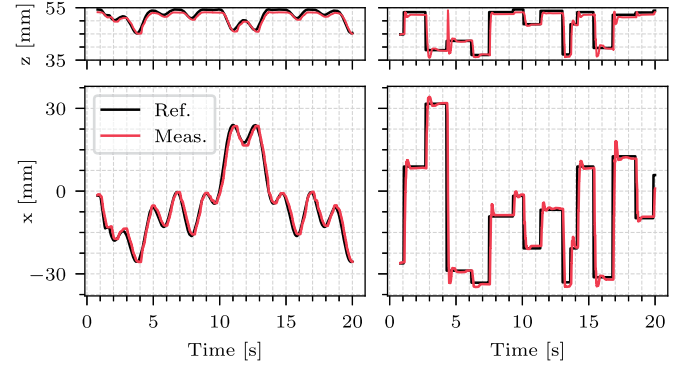


Fig. 10. Closed-loop control experiment results. Left: multi-sinusoidal with frequencies ranging from 0.14 to 0.5 Hz. Right: pseudo-random steps.

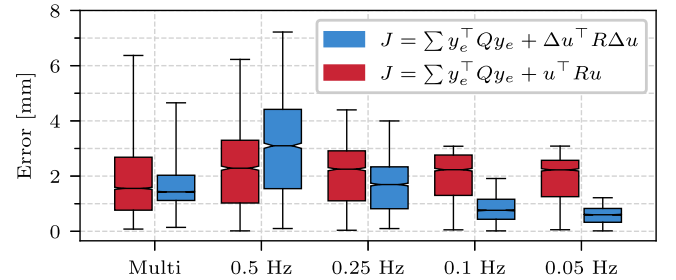


Fig. 11. Closed-loop control errors. Blue represents the error for the proposed control with penalization of the control increment. Red represents the error for penalization of the control signal in the LQ cost function.

used as reference trajectories: (1) a multi-sinusoidal reference signal with frequencies ranging from 0.14 to 0.5 Hz, and (2) a pseudo-random step signal with durations ranging from 0.5 to 2.0 seconds. Throughout these experiments, the lifted state weights Q_φ in the LQ cost function (25) were set to the identity matrix. The input increment weight R was tuned to 0.15. The value of R determines the aggressiveness of the controller. Lower values resulted in large overshoots and oscillations, while higher values slowed down the convergence. The results are shown in Fig. 10. As can be seen, there is no steady-state tracking error. The errors were measured to be 0.60 mm (SD = 0.30) for 0.05 Hz sinusoidal and 1.64 mm (SD = 0.74) for the multi-sinusoidal trajectories. These results are consistent with clinical requirements [33], [34] and are also in line with what has been achieved in the literature for kinematic control, with accuracies of 1.6 mm and 2.09 mm reported in [5] and [35], respectively.

Fig. 11 shows the distribution of the closed-loop control error for the proposed controller, alongside results from implementing conventional LQ controller with penalization of the control signal. The results correspond to the multi-sinusoidal trajectory shown in Fig. 10 and four supplementary sinusoidal reference trajectories. Notably, the proposed controller demonstrated better control performance overall than conventional LQ controller, particularly at lower frequencies of the target signal. The larger error in conventional LQ controller is attributed to steady-state error resulting from the lack of integral action. The submillimeter average error achieved for the sinusoidal target at 0.05 Hz is constrained by the accuracy of the EM tracker. The depreciated tracking performance at higher frequencies is due to

delays introduced by factors such as communication delays, EM tracker sampling time, and the impacts of lower-level control constraints on the physical system. These delays restrict the feasible range for tuning R , thereby exacerbating the tracking error that becomes prominent at higher frequencies.

VII. CONCLUSION

In this work, we proposed a continuous spectrum deep Koopman (CSDK) approach for modeling the dynamics of tendon-driven continuum robots (TDCRs). We augmented the model with two auxiliary networks to estimate the state-dependent parameters of the deep Koopman model and trained the networks on a dataset generated using a calibrated Cosserat rod model. The proposed model was used to implement an LQ controller, resulting in a computationally efficient closed-loop control and high experimental accuracy, demonstrating its potential for implementation across a broad range of applications involving TDCRs. Despite the advancements presented in this work, some limitations exist. Most notably, the inability to capture the frictional effects is due to the use of the frictionless Cosserat rod model for training data generation. Our future research will focus on addressing this limitation, as well as implementing a robust controller, and extending the proposed method to account for out-of-plane motions of the catheter.

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